Assigment - 1

2/11/26

① Design a DFA to accept strings that start and end with same symbols over $\Sigma = \{0, 1\}$. Write the formal definition of DFA. Also show that string 0101100 is accepted by DFA.

Answer

Define States

q_0 - Initial State

q_1 - First Symbol is 0, current last is 0

q_2 - First Symbol is 1, current last is

q_3 - First Symbol is 0, current last is 1

q_4 - First Symbol is 1, current last is 0

Transition table :

Present table	Input 0	Inp
q_0	q_1	q_2
q_1	q_1	q_3
q_2	q_4	q_2
q_3	q_1	q_3
q_4	q_4	q_2

Initial = q_0 , Final States = q_1, q_2

DFA Formal definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

Input String : 010100

Input	Start transition
Start	q_0
0	
1	q_1
0	q_3
1	
0	q_1
0	q_3
	q_1
	q_2

Final State = q_1

String is accepted.

(2) Design NFA to accept string having 101 as Substring.

Answer

States :

q_0 - Initial State

q_1 - First Symbol 1 is detected

q_2 - Substring 10 detected

q_3 - Substring 101 detected (final state)

Transition table :

State	Input 0	Input 1
$\rightarrow q_0$		
q_1		ϕ
q_2	ϕ	
$*q_3$		

Formal definition :

$$M = (Q, \Sigma, \delta, q_0, F)$$

State transitions :

$$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$$

Input Strings :

1101010

Path => $q_0 \to q_1 \to q_3 \to q_3$

String is accepted

NFA accepts all String Containing 101 as substring.

(3) Apply the Principle of mathematical induction and prove that for all positive integers n the statement given below is true.

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Answer

Base case (n=1)

LHS :

$$\sum_{i=1}^{1} i(i+1) = 1(1+1) = 2$$

R.H.S :

$$= \frac{1(1+1)(1+2)}{3} = \frac{6}{3}$$

$$= 2$$

Induction hypothesis :

Assume that Statement is true for $n=k$

$$\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$$

To Prove : $n = k+1$

L.H.S

$$\sum_{i=1}^{k+1} i(i+1)$$

$$= \sum_{i=1}^{k} i(i+1) + (k+1)(k+2)$$

using induction hypothesis,

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= (k+1)(k+2)\left(\frac{k}{3}+1\right)$$

$$= \frac{(k+1)(k+2)(k+3)}{3} = \text{R.H.S}$$

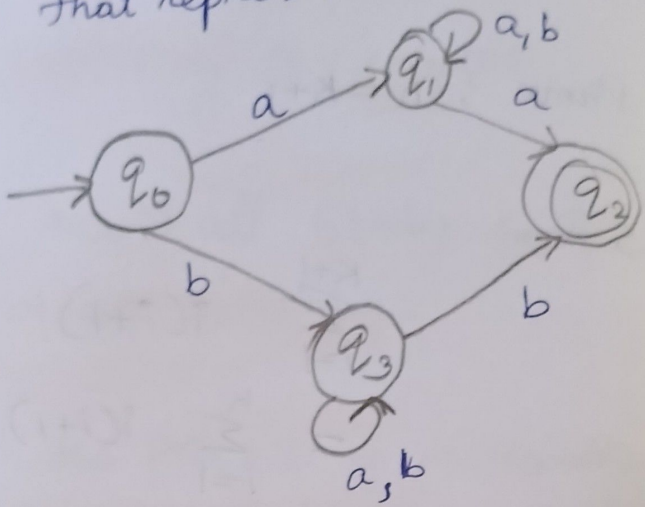
L.H.S = R.H.S

Therefore by principle of mathematical induction,

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Hence, Proved!

(14) consider NFA given below which accept the language representing the set of all strings that start & end same sym over alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents same language.



Answer

$$\Sigma = \{a, b\}$$

NFA States :

q_0 – Initial States

q_1 – State reached after reading

q_3 – State reached after reading

q_2 – Final State

NFA transitions,

State	Input a	Input b
$\rightarrow q_0$		
q_1		
q_3		
$*q_2$	$\emptyset$	$\emptyset$

DFA transition table,

DFA State	Input a	Input b
→A=	B	C
B=	D	B
C=	C	E
*D=	D	B
*E=	C	E

Final States

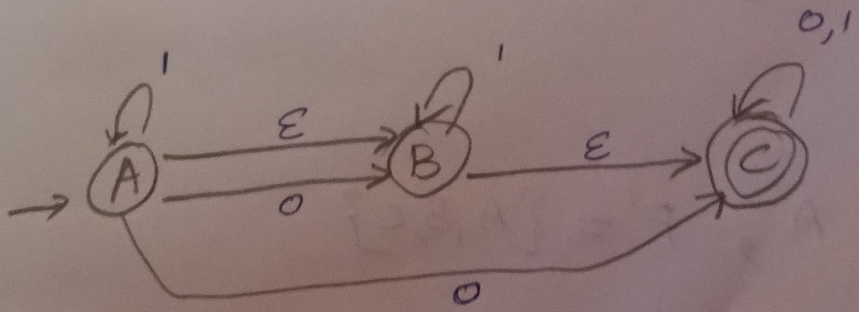
$$F' = \{D, E\}$$

Initial State

$$A = \{q_0\}$$

Equivalent DFA is obtained Successfully

(5) A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent NFA without ε-transitions that accepts the same language. Draw transition table and mark Initial & Final States.



Answer

A - Initial State
B - Intermediate State
C - Final State

$$\Sigma = \{0, 1\}$$

ε-closure(A) :

$$\varepsilon\text{-closure}(A) = \{A, B, C\}$$

ε-closure(B) :

$$\varepsilon\text{-closure}(B) = \{B, C\}$$

ε-closure(C) :

$$\varepsilon\text{-closure}(C) = \{C\}$$

Transition table :

State	Input 0	Input 1
A		
B		
C		

$$q_0 = A, \quad F' = \{A, B, C\}$$

The equivalent NFA without ε-transitions accepts same language as original NFA.